Problem Set 9 - Solutions

Problem 5.43

Heat in the amount of 150 kJ is transferred directly from a hot reservoir at $T_H = 550\,K$ to two cooler reservoirs at $T_1 = 350\,K$ and $T_2 = 250\,K$. The surroundings temperature is $T_\sigma = 300\,K$. If the heat transferred to the reservoir at T_1 is half that transferred to the reservoir at T_2, calculate:

(a) The entropy generation in kJ/K.
(b) The lost work.
(c) How could the process be made reversible?

Solution - Part (a)

In[1]:=
```
(*hot reservoir*)
Q_H = -150.;(*kJ*)
T_H = 550;(*K*)
S_H = Q_H / T_H;(*kJ/kg*)

(*cold reservoir 1 at 350K*)
Q_1 = 50;
T_1 = 350;
S_1 = Q_1 / T_1;

(*cold reservoir 2 at 250K*)
Q_2 = 100;
T_2 = 250;
S_2 = Q_2 / T_2;
```

In[10]:=
```
S_G = S_H + S_1 + S_2 (*kJ/kg*)
```

Out[10]=
```
0.270 129 9
```

The total entropy generation is 0.27013 $\frac{kJ}{K}$. //ANS

Solution - Part (b)

Use equation 5.29:

In[11]:=
```
T_σ = 300;
W_lost = T_σ * S_G (*kJ*)
```

Out[12]=
```
81.03896
```

The lost work is 81.039 kJ. //ANS

Solution - Part (c)

By the second law, the process can be made to be reversible if the total entropy generation is somehow made to be zero. This means that an entropy term must be added that compensates for the total entropy generated. One possibility is to couple some sort of additional reservoir that could remove entropy by heat transfer. Another possibility would be to add some kind of chemical reaction to consume entropy. //ANS

In[•]:=
```
eq1 = 0 == Q_H/T_H + Q_1/T_1 + Q_2/T_2 + s
```

Out[•]=
```
0 == 0.270129 + s
```

In[•]:=
```
Solve[eq1, s]
```

Out[•]=
```
{{s → -0.270129}}
```

Adding a device or chemical reaction with an entropy change of -0.27013 $\frac{kJ}{K}$ of entropy will make the total entropy change zero. //ANS

Problem 5.44

A nuclear power plant generates 750 MW. The reactor temperature is 315°C and a river with water temperature of 20°C is available.

(a) What is the maximum possible thermal efficiency of the plant, and what is the minimum rate at which heat must be discarded to the river?

(b) If the actual thermal efficiency of the plant is 60% of the maximum, at what rate must heat be discarded to the river, and what is the temperature rise of the river if it has a flow rate of 165 cubic meters per second?

Solution - Part (a)

In[18]:=
```
T_C = 273.15 + 20; (*river temperature in K*)
T_H = 273.15 + 315; (*reactor temperature in K*)
η_max = 1 - T_C/T_H (*eq. 5.7*)
```

Out[20]=
```
0.501572 7
```

The maximum possible thermal efficiency η_{max} is 0.501<u>6</u>. //ANS

In[21]:=
```
Q_H = 750 / η_max (*eq. 5.6*)
```

Out[21]=
```
1495.297
```

In[22]:=
```
Qc = QH - 750
```

Out[22]=
```
745.2966
```

A minimum of 745.3 MW of heat must be discarded to the river. //ANS

Solution - Part (b)

Recalculate Q_C using the new efficiency, then use $Q_C = \dot{m}\,Cp\,\Delta T$ with the given flow rate.

In[23]:=
```
ηactual = .60 * ηmax;
QH = 750 / ηactual;
```

In[25]:=
```
Qc = QH - 750
```

Out[25]=
```
1742.161
```

At least 1742.2 MW of heat are discarded to the river. //ANS

Now use $Q_C = \dot{m}\,Cp\,\Delta T$, where mass flow rate $\dot{m}$ is obtained from the given volumetric flow rate. Density is about 1 kg/L and specific heat is 4.184 kJ/kg·°C, both lookups from google.

In[26]:=
```
eq1 = 165 m^3/s * 1000 L/m^3 * 1 kg/L * 4.184 kJ/(kg * °C) * ΔT == 1742.16 MW * 1000 kW/MW * 1 kJ/s; (*Qc=ṁCpΔT*)
```

In[27]:=
```
Solve[eq1, ΔT]
```

Out[27]=
```
{{ΔT → 2.523553 °C}}
```

The temperature increase of the river is 2.52°C. //ANS

Problem 5.50

Ethylene vapor is cooled at atmospheric pressure from 830 to 35°C by direct heat transfer to the surroundings at 25°C. With respect to this surroundings temperature, what is the lost work of the process in kJ/mol?

Show that the same result is obtained as the work which can be derived from reversible heat engines operating with the ethylene vapor as the heat source and the surroundings as the sink. The heat capacity of ethylene is given in Table C.1 of App. C.

Solution - Part (a)

This problem involves integrating the heat capacity polynomial. Recall that temperature must be in K and heat capacity in Appendix C is dimensionless.

Gather information:

In[28]:=
```
Clear[Subscript]
```

In[29]:=
```
T1 = 273.15 + 830; (*initial ethylene temperature (830°C)*)
T2 = 273.15 + 35;(*final ethylene temperature (35°C)*)
Tσ = 273.15 + 25;(*surroundings temperature (25°C)*)

(*gas constant, needed for Cp*)
R = 8.314;(*J/(mol*K)*)
(*Cp coefficients from App. C*)
a = 1.424;
b = 14.394 * 10^-3;
c = -4.392 * 10^-6;
Cp = a + b * T + c * T^2;
```

Use equation 5.31 ($W_{lost} = T_\sigma * \Delta S_{fs} - Q$) with ethylene in the flowing stream (fs) so that ΔS_{fs} refers to the entropy change of the ethylene. Q refers to the heat removed from the ethylene, and T_σ is the temperature of the surroundings.

The temperature of ethylene is changing, so both entropy ΔS_{fs} and Q are calculated by integrating. Calculate ΔS_{fs} by integrating $\frac{C_P}{T}$ with respect to T. Calculate Q by assuming constant pressure so that Q = ΔH with ΔH obtained integrating C_P with respect to T.

In[37]:=
```
ΔSfs = R * ∫_T1^T2 Cp/T dT (*in units of J/(mol*K)*)
```

Out[37]=
```
-89.75325
```

In[38]:=
```
Qethylene = R * ∫_T1^T2 Cp dT (* Q=ΔH at constant pressure, in units of J/mol*)
```

Out[38]=
```
-60 563.01
```

In[39]:=
```
Wlost = Tσ * ΔSfs - Qethylene (*equation 5.31*)
```

Out[39]=
```
33 803.08
```

The lost work is 33.8031 $\frac{kJ}{mol}$. //ANS

Solution - Part (b)

To analyze this process as a reversible engine with the Carnot equations, we need values for Q_H and Q_C. It should be clear that the ethylene vapor is the heat source. That is, the heat added to the engine (the system) comes from the ethylene. This is Q_H in the reversible (Carnot) cycle, and the value is the same as it was before, but opposite in sign because it enters the engine.

In[40]:=
```
QH = -Qethylene (*Heat transferred (QH) is positive with respect to the engine.*)
```

Out[40]=
```
60 563.01
```

Heat transferred (Q_H) is positive with respect to the engine.

To obtain Q_C, we need to understand a couple of things. First, equation 5.31 (shown below) gives the lost work in terms of entropy change of the flowing streams and Q dissipated to the surroundings.

Second, the lost work is zero for a reversible process. Since this Q is heat transferred to the surroundings, it is also Q_C in the reversible (Carnot) cycle. Because the temperature change of the ethylene is the same as it was in part (a), the entropy change of the ethylene is also the same. So we calculate the Q that zeros the lost work in equation 5.31 and make this equal to Q_C in the Carnot engine.

In Equation 5.31, $W_{lost} = T_\sigma \Delta S - Q$, where $W_{lost} = 0$, $\Delta S = \Delta S_{fs}$, and $Q = Q_c$ for a reversible process.

In[41]:=
```
Qc = SolveValues[0 == Tσ * ΔSfs - Qc , Qc][[1]] (*J/mol*)
```

Out[41]=

$$-26\,759.93$$

Heat transferred is negative with respect to the system.

Finally, use the first law ΔU=Q+W (equation 2.3), where ΔU=0 for a cyclic process, and where $Q=Q_H+Q_C$ as explained on page 181. This gives $W=-Q_H-Q_C$.

In[42]:=
```
Wheatengine = -QH - Qc (*J/mol*)
```

Out[42]=

$$-33\,803.08$$

The work derived from reversible heat engine is -33.8031 $\frac{kJ}{mol}$.

The sign is negative because work is leaving the engine (system). The absolute value of the work is the same value as obtained in part (a), that is, |W| = 33.8031 $\frac{kJ}{mol}$. //ANS

Problem 5.17

A Carnot engine operates between temperature levels of 600 K and 300 K. It drives a Carnot refrigerator, which provides cooling at 250 K and discards heat at 300 K. Determine a numerical value for the ratio of heat extracted by the refrigerator ("cooling load") to the heat delivered to the engine ("heating load").

Solution

Calculate the efficiency and work of the Carnot engine:

In[44]:=
```
η = 1 - 300/600 ; (*η=1-Tc/TH*)
```

In[45]:=
```
WE = -QH * η
```

Out[45]=

$$-\frac{Q_H}{2}$$

Calculate the COP and work of the Carnot refrigerator (Lesson 26):

In[46]:=
```
COP = 250/(300 - 250) ; (*COP=Tc/(TH-Tc)*)
```

In[47]:=
```
W_R = +Q_C / COP
```

Out[47]=

$$\frac{Q_C}{5}$$

The work entering the refrigerator is equal to but opposite in sign to the work leaving the engine:

In[48]:=
```
W_R == -W_E
```

Out[48]=

$$\frac{Q_C}{5} == \frac{Q_H}{2}$$

In[52]:=
```
Print["Q_C / Q_H =", (Q_C / Q_H) /. Solve[W_R == -W_E, Q_C][[1]]]
```

$$\frac{Q_C}{Q_H} = \frac{5}{2}$$

The ratio of the cooling load to the heating load is 5/2. In other words, $\frac{Q_C}{Q_H} = \frac{5}{2}$. //ANS